

CHALLENGE

Towards reliable compound screening

By combining uncertainty quantification with active learning...

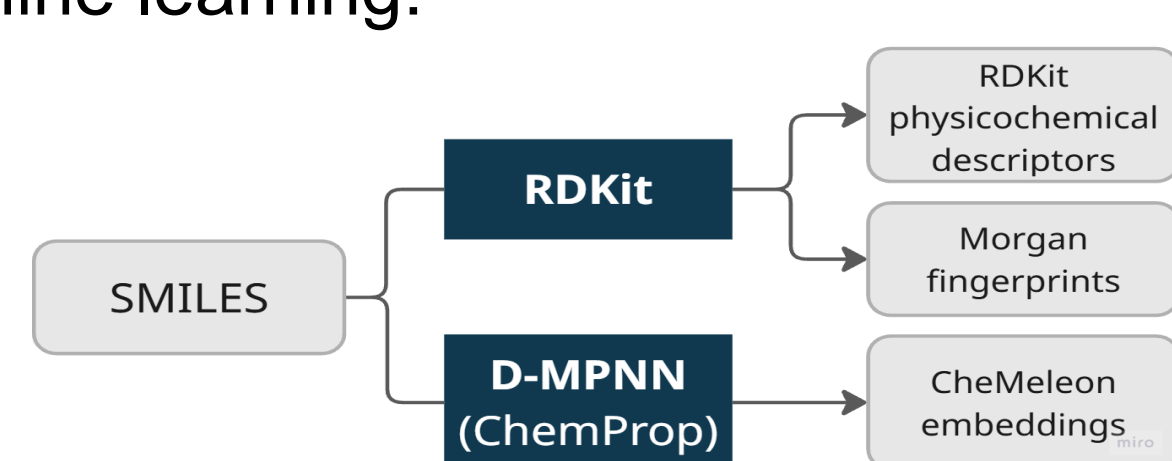
- computational workflows for molecular discovery become more robust
- experimental resources can be focused on the most informative compounds
- confident data-driven candidate selection is enabled

KEY CONCEPTS

Representing and selecting molecular compounds

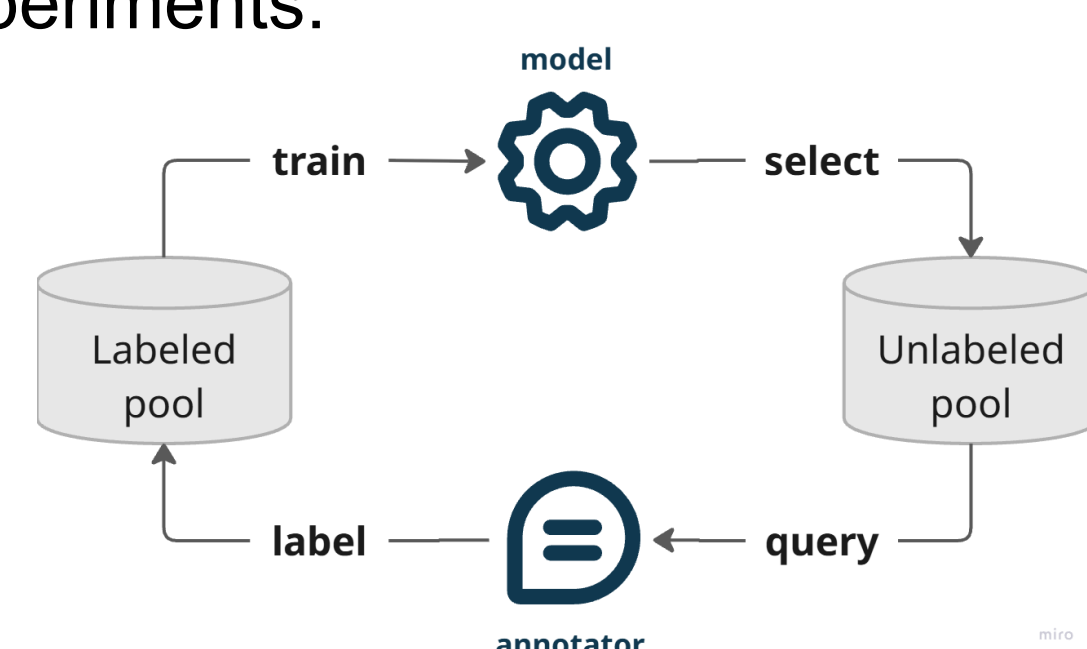
1 How to **represent molecules** for machine learning:

Different feature sets are available for tabular representation of molecular compounds. Canonical SMILES serve as a basis for feature extraction.



2 Using **active learning (AL)** to guide experiments:

AL increases data efficiency by iteratively selecting the most informative samples for labeling based on a small labeled dataset. Across multiple iterations, the labeled dataset grows.



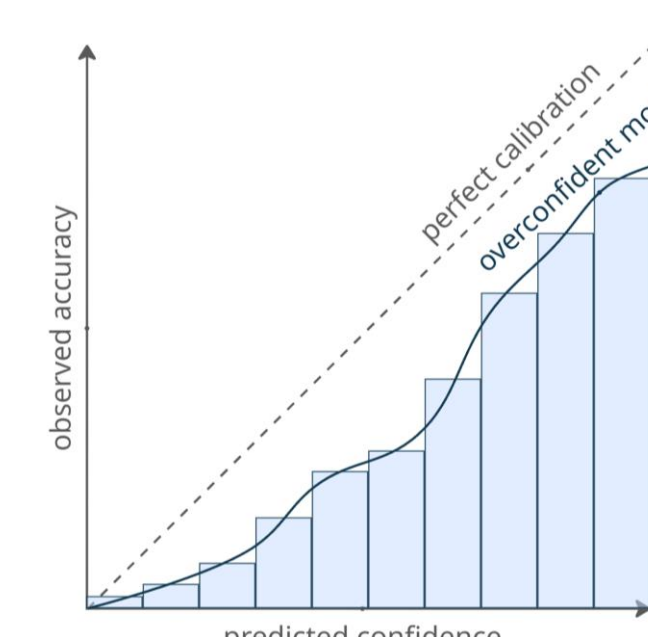
APPROACH

Uncertainty-aware active learning framework

Molecular prediction benchmarks were evaluated across multiple bioactivity datasets [1], molecular representations, and prediction models. Performance was assessed using predictive accuracy and uncertainty quality. Finally, a reusable framework for uncertainty-aware AL was built.

Uncertainty quantification metrics

- Predicted confidence
- Entropy-based uncertainty
- Ensemble disagreement score [2]
- Calibration metrics:
 - Expected calibration error (ECE) [3]
 - Brier Score [4]



Molecular prediction models

TabPFN [5, 6] transformer-based foundation model

CatBoost [7] tree-based gradient boosting

Neural network (NN) deep learning baseline

In addition, **hybrid models** were constructed from the base models for AL evaluation. Each hybrid model consists of an embedding extractor and a prediction head.

Active learning query methods

Baseline random acquisition

Uncertainty only entropy, BALD [8], Bregman Information [9], CatBoost [10]

Uncertainty + diversity BADGE [11], custom diversity method (HDBSCAN [12])

KEY FINDINGS

Providing reliable predictions

- Predictive performance** is largely independent on model architecture and molecular representation. Models achieve balanced accuracies of 60-90%.
- Model and feature representation strongly influence **calibration** quality for all included datasets

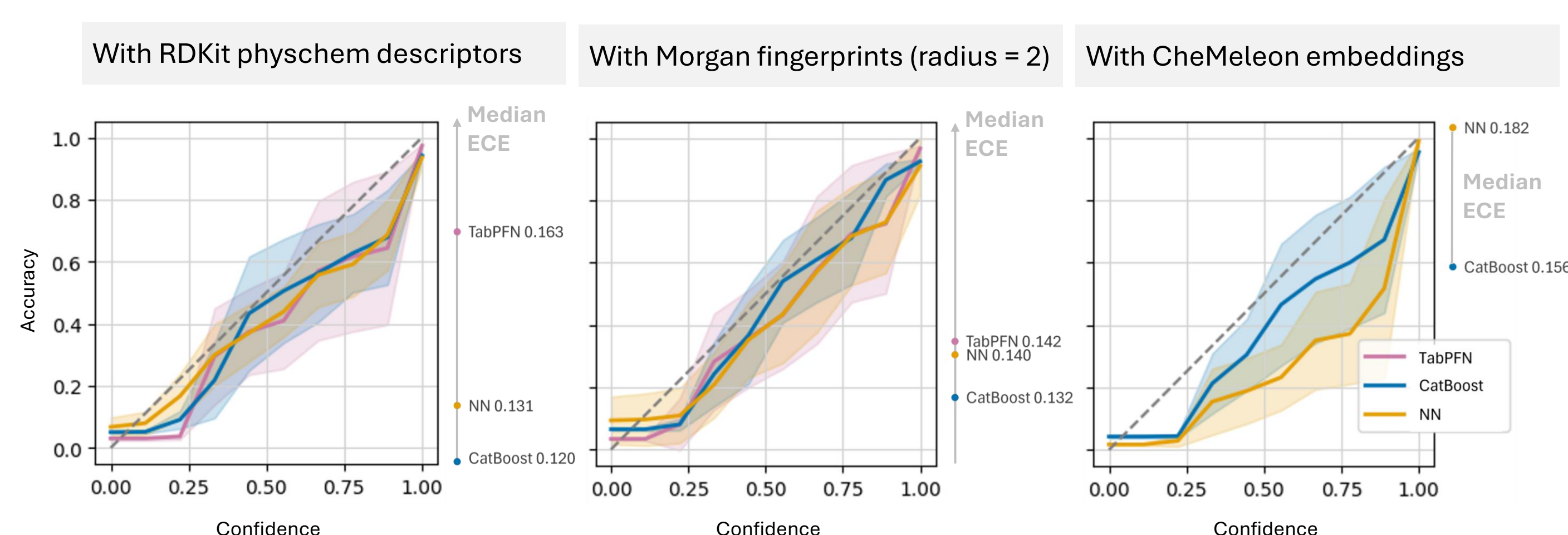


Fig. 1: Calibration curves for the main dataset including standard deviation across repeated runs. Median ECE across repeated runs is displayed on an additional axis.

Accelerating efficient learning

- Uncertainty-guided acquisition improves **active learning efficiency**
- Greatest benefits achieved when uncertainty and diversity are combined
- TabPFN and CatBoost exhibit more robust learning behavior than neural networks

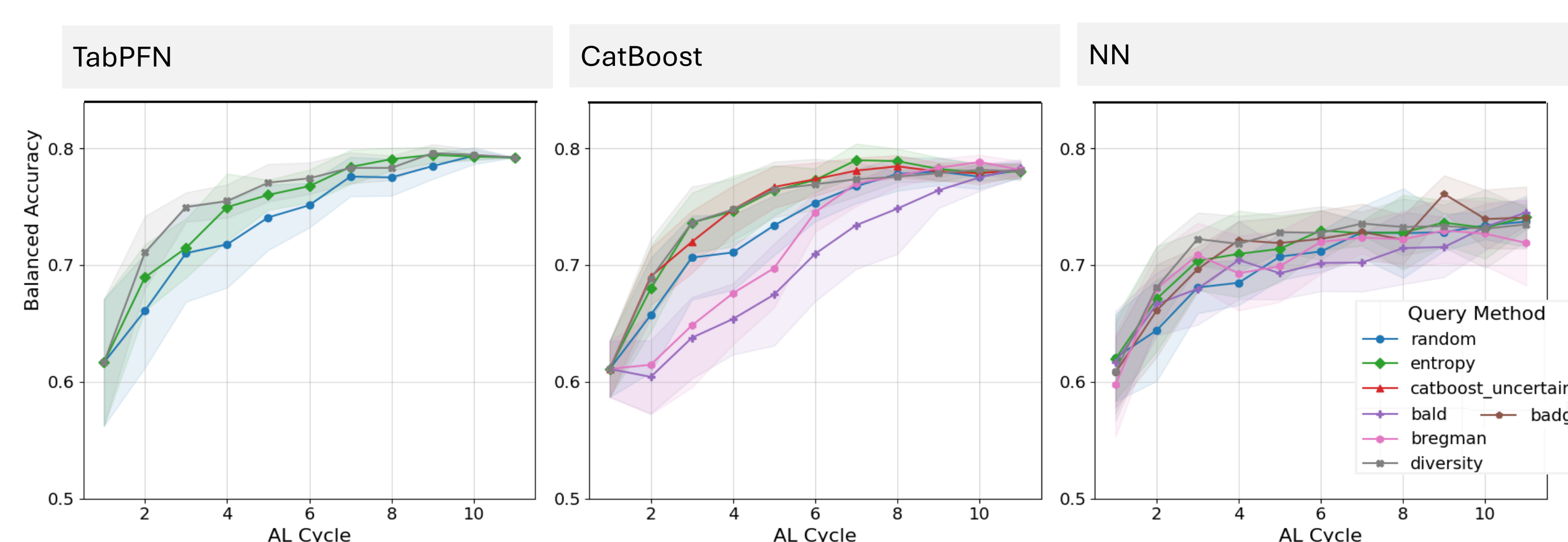


Fig. 2: Mean learning curves with standard deviation across repeated runs for different models and query methods. The random baseline method is depicted in blue.

OUTLOOK

From research to practice

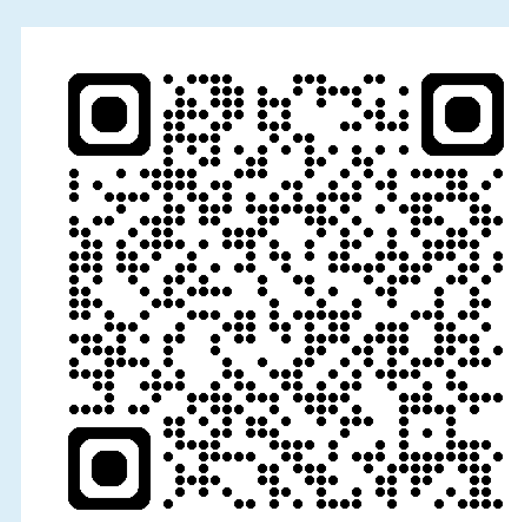
Current work at Bayer explores how uncertainty-aware machine learning can be integrated into real-world discovery workflows.

- Integration of uncertainty metrics and AL methods into MotherML, Bayer's open-source machine learning framework
- Enabling trustworthy model predictions at scale
- Supporting efficient compound prioritization and screening campaigns

Let's connect! Scan to explore the full project:

- Poster PDF
- Full Master's thesis
- Bayer's MotherML framework
- MLO Lab website (DKFZ)
- SysMed website (University of Augsburg)

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<https://junkha.github.io/active-learning-poster-resources/>

References

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- [2] Lakshminarayanan et al., Deep Ensembles, NIPS 2017
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- [4] Brier, Brier Score, 1950
- [5] Hollmann et al., TabPFN, Nature 2025
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